

If (at $\rho \in \mathcal{P}_2^r$, $\partial_{\mathcal{W}_2} F(\rho)$ is non empty and) $\xi \in \partial_{\mathcal{W}_2} F(\rho)$, then

$$\xi + \frac{T_\rho^p - \text{Id}}{\gamma} \in \partial_{\mathcal{W}_2} G(\rho).$$

The argument follows that in Lemma 10.1.2 in [7], and we include a proof for completeness.

Proof of Lemma A.1. We are to verify that

$$\eta := \xi + \frac{T_\rho^p - \text{Id}}{\gamma}$$

is a strong subdifferential of G at ρ . By Definition 2.2, it suffices to show that for any $v \in L^2(\rho)$ and $\delta \rightarrow 0$,

$$G((\text{Id} + \delta v)_\# \rho) - G(\rho) \geq \delta \langle \eta, v \rangle_\rho + o(\delta). \quad (\text{A.2})$$

By construction,

$$\langle \eta, v \rangle_\rho = \langle \xi, v \rangle_\rho + \frac{1}{\gamma} \langle T_\rho^p - \text{Id}, v \rangle_\rho,$$

and since ξ is a strong subdifferential of F at ρ ,

$$F((\text{Id} + \delta v)_\# \rho) - F(\rho) \geq \delta \langle \xi, v \rangle_\rho + o(\delta).$$

Combining the two and by the definition of F , we can deduce (A.2) as long as we can show that

$$\frac{1}{2} \mathcal{W}_2(p, (\text{Id} + \delta v)_\# \rho)^2 + o(\delta) \leq \frac{1}{2} \mathcal{W}_2(p, \rho)^2 - \langle T_\rho^p - \text{Id}, \delta v \rangle_\rho. \quad (\text{A.3})$$

To show (A.3), note that by Brenier Theorem (ii),

$$\mathcal{W}_2(p, \rho)^2 = \int \|x - T_\rho^p(x)\|^2 \rho(x) dx = \|\text{Id} - T_\rho^p\|_\rho^2.$$

Thus,

$$\begin{aligned} & \frac{1}{2} \mathcal{W}_2(p, \rho)^2 - \langle T_\rho^p - \text{Id}, \delta v \rangle_\rho \\ &= \frac{1}{2} \|\text{Id} - T_\rho^p\|_\rho^2 + \langle \text{Id} - T_\rho^p, \delta v \rangle_\rho \\ &= \frac{1}{2} \|(\text{Id} + \delta v) - T_\rho^p\|_\rho^2 - \frac{1}{2} \|\delta v\|_\rho^2. \end{aligned} \quad (\text{A.4})$$

Note that, because $v \in L^2(\rho)$,

$$\|\delta v\|_\rho^2 = O(\delta^2)$$

and

$$\begin{aligned} \|(\text{Id} + \delta v) - T_\rho^p\|_\rho^2 &= \int_{\mathbb{R}^d} \|(\text{Id} + \delta v)(x) - T_\rho^p(x)\|^2 \rho(x) dx \\ &\geq \mathcal{W}_2((\text{Id} + \delta v)_\# \rho, p)^2. \end{aligned}$$

Putting together, this gives that the r.h.s. of (A.4) is greater than or equal to

$$\frac{1}{2} \mathcal{W}_2((\text{Id} + \delta v)_\# \rho, p)^2 + O(\delta^2)$$

which implies (A.3). □

A.2 Proofs of Lemmas

Proof of Lemma 3.2. It suffices to show that for any A s.t. $\text{Leb}(A) = 0$, $T_{\#}P(A) = 0$. By the definition of push-forward, $T_{\#}P(A) = P(T^{-1}(A))$, which is zero because $\text{Leb}(T^{-1}(A)) = 0$ (since T is non-degenerate) and $P \ll \text{Leb}$. $\square$

Proof of Lemma 3.3. First, the minimizer makes the r.h.s. finite because $T = \text{Id}$ makes it finite: When T is identity, the r.h.s. equals $G(p_n) < \infty$. As a result, $\tilde{p} := T_{\#}p_n$ needs to have density because otherwise the KL divergence $G(\tilde{p}) = +\infty$.

It remains to show that $M_2(\tilde{p}) < \infty$. By definition,

$$\begin{aligned} M_2(\tilde{p}) &= \int_{\mathbb{R}^d} \|x\|^2 \tilde{p}(x) dx \\ &= \mathbb{E}_{x \sim p_n} \|T(x)\|^2 \\ &\leq 2(\mathbb{E}_{x \sim p_n} \|x\|^2 + \mathbb{E}_{x \sim p_n} \|x - T(x)\|^2), \end{aligned}$$

where $\mathbb{E}_{x \sim p_n} \|x\|^2 = M_2(p_n) < \infty$, and, at the minimizer T , $\mathbb{E}_{x \sim p_n} \|x - T(x)\|^2$ also needs to be finite due to that it is in the 2nd term of (25). $\square$

Proof of Lemma 3.4. We first show that $T \in L^2(p)$ iff $T_{\#}p \in \mathcal{P}_2$. This is because $M_2(T_{\#}p) = \mathbb{E}_{x \sim p} \|T(x)\|^2$, and thus is finite iff. T is in $L^2(p)$.

As a result, when $T \in L^2(p)$, $T_{\#}p \in \mathcal{P}_2$. If T is also non-degenerate, Lemma 3.2 implies that $T_{\#}p$ also has density. This proves that $T_{\#}p \in \mathcal{P}_2^r$. $\square$

B Proofs and lemmas in Section 4

B.1 Technical lemmas in Section 4.1

Lemma B.1. $\mathcal{H}(\rho)$ is convex a.g.g. in $\mathcal{P}_2$.

Proof. The a.g.g.-convexity of functional in the form of $\mathcal{F}(\rho) = \int F(\rho(x))dx$ in $\mathcal{P}_2$ is established in Proposition 9.3.9 of [7] when $F : [0, +\infty) \rightarrow (-\infty, \infty]$ is a proper, lower semi-continuous convex function satisfying that $s \mapsto s^d F(s^{-d})$ is convex and non-increasing on $(0, +\infty)$. The entropy $\mathcal{H}(\rho) = \mathcal{F}(\rho)$ with $F(s) = s \log s$, and this F satisfies the above conditions. $\square$

Lemma B.2. Under Assumption 4, $\mathcal{E}(\rho)$ is λ -convex a.g.g. in $\mathcal{P}_2$.

Proof. This is a direct result of Proposition 9.3.2(i) of [7], noting that assuming the boundedness of V^- implies the growth condition needed in Section 9.3 therein. The proof of Proposition 9.3.2(i) shows that $\mathcal{E}(\rho)$ is λ -convex along any interpolation curve which implies λ -convexity a.g.g. $\square$

Lemma B.3. Under Assumptions 4, the KL divergence $G(\rho)$ defined in (16) satisfies Assumption 3.

Proof. The lower semi-continuity follows from that of $\mathcal{H}(\rho)$ and the condition on V in Assumption 4. The domain of G is restricted to ρ with density because $\mathcal{H}(\rho)$ diverges otherwise. The a.g.g. λ -convexity of G directly follows from Lemma B.1 and Lemma B.2. $\square$

B.2 Proofs in Section 4.2

Proof of Lemma 4.1. The unique existences of T_p^ρ and T_p^π are by Brenier Theorem. Since $\rho \in \mathcal{P}_2^r$, the map T_p^ρ has an inverse denoted by T_ρ^p which is defined ρ -a.e. Under Assumption 3 first part, the strong subdifferential of $\partial_{\mathcal{W}_2} G(\rho)$ is well-defined, and we assume η is one of them.

Let $v := T_p^\pi \circ T_\rho^p - \text{Id}$. One can verify that $v \in L^2(\rho)$, since $\|T_p^\pi \circ T_\rho^p\|_\rho^2 = M_2(\pi)$, $\|\text{Id}\|_\rho^2 = M_2(\rho)$, and both are finite. By definition, for $\delta \in [0, 1]$,

$$\begin{aligned} (\text{Id} + \delta v)_\# \rho &= (\text{Id} + \delta(T_p^\pi \circ T_\rho^p - \text{Id}))_\# \rho \\ &= (T_p^\rho + \delta(T_p^\pi - T_p^\rho))_\# p. \end{aligned} \quad (\text{A.5})$$

We also have

$$\langle \eta, v \rangle_\rho = \langle \eta \circ T_p^\rho, T_p^\pi - T_p^\rho \rangle_p. \quad (\text{A.6})$$

Since $v \in L^2(\rho)$, by that $\eta \in \partial_{\mathcal{W}_2} G(\rho)$ and the definition of strong subdifferential (Definition 2.2), with $\delta \rightarrow 0+$ we have

$$G((\text{Id} + \delta v)_\# \rho) \geq G(\rho) + \delta \langle \eta, v \rangle_\rho + o(\delta). \quad (\text{A.7})$$

Combined with (A.5)(A.6), this gives

$$\begin{aligned} &G((T_p^\rho + \delta(T_p^\pi - T_p^\rho))_\# p) - G(\rho) \\ &\geq \delta \langle \eta \circ T_p^\rho, T_p^\pi - T_p^\rho \rangle_p + o(\delta). \end{aligned} \quad (\text{A.8})$$

Meanwhile, by the λ -convexity of G a.g.g. (Definition 2.3), and specifically (14), we have

$$\begin{aligned} &G((T_p^\rho + \delta(T_p^\pi - T_p^\rho))_\# p) \\ &\leq (1 - \delta)G(\rho) + \delta G(\pi) - \frac{\lambda}{2} \delta(1 - \delta) \mathcal{W}_2(\rho, \pi)^2. \end{aligned} \quad (\text{A.9})$$

Comparing (A.8) and (A.9), we have

$$\begin{aligned} G(\pi) - G(\rho) &\geq \langle \eta \circ T_p^\rho, T_p^\pi - T_p^\rho \rangle_p \\ &\quad + \frac{\lambda}{2} (1 - \delta) \mathcal{W}_2(\rho, \pi)^2 + o(1). \end{aligned}$$

We get the conclusion by letting $\delta \rightarrow 0+$. □

Proof of Lemma 4.2. In the n -th step, T_{n+1} is in $L^2(p_n)$ and is non-degenerate under Assumption 1, thus from $p_n \in \mathcal{P}_2^r$, $p_{n+1} = (T_{n+1})_\# p_n$ is also in $\mathcal{P}_2^r$ by Lemma 3.4. This holds for $n = 0, \dots, N-1$, and thus all p_n are in $\mathcal{P}_2^r$, where $p_0 \in \mathcal{P}_2^r$ is by the lemma assumption. By the Brenier Theorem, the OT map from p_n to p_{n+1} is denoted as T_n^{n+1} , which is uniquely defined p_n -a.e. Let T_{n+1}^n be the OT map from p_{n+1} to p_n , and it is also the p_{n+1} -a.e. inverse of T_n^{n+1} . We use the short-hand notation

$$X_{n+1} := T_n^{n+1}.$$

Under the assumption on G , Lemma A.1 applies which gives the relationship between $\partial_{\mathcal{W}_2} F_{n+1}$ and $\partial_{\mathcal{W}_2} G$. Together with the assumption on ξ_{n+1} by Assumption 1, we have that for each n , $\exists \eta_{n+1} \in \partial_{\mathcal{W}_2} G(p_{n+1})$ s.t.

$$\gamma \xi_{n+1} - \gamma \eta_{n+1} = \text{Id} - T_{n+1}^n, \quad p_{n+1}\text{-a.e.}$$